MASTER DOCUMENT TREE

250 Engineering Documents

Ultra-Safe Central Capsule – Large Configuration

Aligned to Document 00 Version 3.0 — Comprehensive Engineering Baseline

5.80 m × 3.80 m | 22 000 kg Hard Ceiling | 8 Landing Legs | Six-Layer Survival Architecture

Document Version	Date	Status	Total Documents
1.1	September 2026	Controlled Tree aligned to Document 00 Version 3.0	**250**

This Master Document Tree defines the complete hierarchical structure of 250 engineering documents required to develop the Ultra-Safe Central Capsule (Large Configuration) under the controlling baseline of Document 00 Version 3.0 — Comprehensive Engineering Baseline. The tree is verification-gated, configuration-controlled and explicitly distinguishes locked baseline values, engineering requirements, assumptions, derived targets and open verification gates. No performance claim is treated as demonstrated unless supported by controlled analysis, test evidence or approved qualification. Documents are grouped by technical domain and numbered for full traceability. Every document title has an associated Document Type.

1. Document Package Overview by Domain

Group	Domain	Doc Range	Count
A	Master Control, Governance & Configuration	001 – 020	20
B	Requirements, Safety Case & Certification Planning	021 – 050	30
C	Systems Engineering, Architecture & Independence	051 – 080	30
D	Structural Design, Roll-Cage & Continuous Floor	081 – 115	35
E	Separation, Six-Layer Safety & Common-Cause Control	116 – 155	40
F	Parachute, Landing Legs & Impact Energy Management	156 – 185	30
G	Life Support, Fire, Flotation & Post-Event Survival	186 – 210	25
H	Avionics, Electrical, Software & Common-Cause Analysis	211 – 235	25
I	Human Factors, Egress, Verification Gates & Traceability	236 – 250	15
	TOTAL	**001 – 250**	**250**

2. Complete Document Tree (001 – 250) with Document Types

All documents are controlled against Document 00 Version 3.0. Titles reflect verification-gated, configuration-controlled engineering content. Locked baseline values (geometry, mass ceiling, 8 legs, continuous floor, six-layer architecture) shall not be silently modified. Performance claims remain verification-gated until controlled evidence exists.

No.	Document Title	Document Type
	GROUP A – Master Control, Governance & Configuration (001 – 020)	
1	Master Document Index & Document Tree (this document)	Management
2	Project Charter & Engineering Authority Statement	Management
3	Scope Definition & System Boundaries (Controlled)	Requirements
4	Configuration Management Plan (aligned to Doc 00 V3.0)	Management
5	Document Control & Change Management Procedure	Management
6	Baseline Control & Locked-Value Register	Management
7	Risk Management Plan & Initial Risk Register	Management
8	Safety Management System Overview	Safety
9	Quality Management Plan	Quality
10	Systems Engineering Management Plan	Management
11	Interface Control Document (ICD) Master Register	ICD
12	Requirements Traceability Matrix (Master)	Management
13	Verification & Validation Master Plan (Verification-Gated)	Test
14	Technology Readiness & Evidence Maturity Assessment	Assessment
15	Make-or-Buy & Supply Chain Strategy	Management
16	Program Schedule, Milestones & Resource Plan	Management
17	Assumption Register & Closure Tracking	Management
18	Open Issues & Verification Gate Register	Management
19	Nonconformance & Corrective Action Process	Quality
20	Engineering-to-IP Traceability Control Plan	Management
	GROUP B – Requirements, Safety Case & Certification Planning (021 – 050)	
21	System Requirements Specification (SRS) – Top Level	Requirements
22	Functional Requirements Specification	Requirements
23	Performance Requirements Specification (Verification-Gated)	Requirements
24	Safety Requirements Specification	Safety
25	Environmental Requirements Specification	Requirements
26	Human Factors Requirements	Requirements
27	Reliability, Availability, Maintainability (RAM) Requirements	Requirements
28	Interface Requirements Specification	Requirements
29	Derived Requirements & Constraints Register	Requirements
30	Requirements Validation Report	Requirements
31	Hazard Identification & Preliminary Hazard Analysis (PHA)	Safety
32	Functional Hazard Assessment (FHA)	Safety
33	System Safety Assessment (SSA) Plan	Safety
34	Common-Cause Analysis (CCA) Plan & Register	Safety
35	Failure Modes, Effects and Criticality Analysis (FMECA) – System Level	Safety
36	Fault Tree Analysis (FTA) – Critical Functions	Safety
37	Event Tree Analysis (ETA)	Safety
38	Safety Case Part 1 – Argument Structure	Safety
39	Safety Case Part 2 – Evidence & Claims (Verification-Gated)	Safety
40	Airworthiness Case Outline (Capsule as Airborne Module)	Certification

No.	Document Title	Document Type
41	Maritime Safety Case Outline (Host Integration)	Certification
42	Certification Basis – Aviation Regulations Mapping	Certification
43	Certification Basis – Maritime Regulations Mapping	Certification
44	Means of Compliance Matrix	Certification
45	Particular Risk Analysis (Fire, Explosion, Impact, Water)	Safety
46	Occupant Safety & Crashworthiness Requirements	Safety
47	Emergency Egress & Rescue Requirements	Requirements
48	Software Safety Assurance Plan Outline	Safety
49	Certification Project Plan (Evidence-Gated)	Certification
50	Post-Certification Change Process	Certification
	GROUP C – Systems Engineering, Architecture & Independence (051 – 080)	
51	System Architecture Description Document	Architecture
52	Functional Architecture	Architecture
53	Physical Architecture & Product Breakdown Structure (PBS)	Architecture
54	Logical Architecture & Data Flow	Architecture
55	Mode & State Definition Document	Architecture
56	Operational Mode Transition Matrix	Architecture
57	System Interface Control Document – Capsule Internal	ICD
58	System ICD – Host Platform Interface (Generic)	ICD
59	Electrical Power Architecture	Architecture
60	Data Architecture & Network Topology	Architecture
61	Redundancy & Fail-Operational Architecture	Architecture
62	Six-Layer Survival Architecture Specification (Controlled)	Architecture
63	Safety-Layer Independence Analysis Plan	Safety
64	Separation Sequence Logic & Timing	Design
65	Deployment Sequence Logic & Timing	Design
66	Recovery & Landing Sequence Logic	Design
67	Human-Machine Interface (HMI) Architecture	Architecture
68	Health Monitoring & Prognostics Architecture	Architecture
69	Configuration Item (CI) Identification & Control	Management
70	System Mass Properties Control Plan (22 000 kg Ceiling)	Management
71	Centre of Gravity & Inertia Management Plan	Analysis
72	Thermal Architecture Overview	Architecture
73	Electromagnetic Compatibility (EMC) Architecture	Architecture
74	Cyber-Physical Security Architecture	Architecture
75	Model-Based Systems Engineering (MBSE) Model Description	Architecture
76	Trade Study Methodology & Controlled Results Log	Analysis
77	System Performance Budgets (Mass, Power, Volume, Timing)	Analysis
78	System Integration Strategy	Management
79	End-to-End Mission Timeline	Operations
80	Open Issues, Assumptions & Verification Gate Register	Management
	GROUP D – Structural Design, Roll-Cage & Continuous Floor (081 – 115)	

No.	Document Title	Document Type
81	Structural Design Philosophy & Criteria (Controlled)	Requirements
82	Primary Structure Design Description – Continuous Roll-Cage	Design
83	Floor Structure Continuous Load Path Design	Design
84	Door & Hatch Structural Reinforcement Design	Design
85	Titanium Node Fitting Design Specification	Specification
86	Landing Leg Attachment Structure Design (8 Legs)	Design
87	Separation Interface Structure Design	Design
88	External Shell / Fairing Structural Design	Design
89	Internal Secondary Structure Design	Design
90	Seat & Restraint Load Path Design	Design
91	Material Selection Report – Carbon Fibre Composites	Materials
92	Material Selection Report – Metals & Fasteners	Materials
93	Material Allowables & Design Values (Source-Controlled)	Materials
94	Joint Design & Bonding Specification	Specification
95	Fastener Selection & Installation Specification	Specification
96	Corrosion Protection & Coatings Specification	Specification
97	Structural Mass Breakdown & Optimization Report	Analysis
98	Finite Element Model Description – Global Model	Analysis
99	Finite Element Model Description – Detailed Sub-models	Analysis
100	Static Load Analysis Report (Verification-Gated)	Analysis
101	Dynamic & Modal Analysis Report	Analysis
102	Impact & Crash Analysis Report (Multi-axis)	Analysis
103	Roll-over & Inversion Analysis Report	Analysis
104	Fatigue & Damage Tolerance Analysis	Analysis
105	Buckling & Stability Analysis	Analysis
106	Thermal Stress Analysis	Analysis
107	Structural Safety Factor Justification (≥ 2.8 Target)	Analysis
108	Damage Tolerance & Fail-Safe Assessment	Safety
109	Residual Strength after Separation / Impact	Analysis
110	Structural Test Requirements Specification	Test
111	Structural Test Plan – Static	Test
112	Structural Test Plan – Dynamic / Impact	Test
113	Structural Design Review Checklist	Review
114	Critical Structural Items List	Management
115	Structural Design Summary & Compliance Matrix	Management
	GROUP E – Separation, Six-Layer Safety & Common-Cause Control (116 – 155)	
116	Separation System Design Philosophy	Architecture
117	Separation Interface Geometry & Clearance	Design
118	Pyrotechnic Separation Actuator Specification	Specification
119	Hydraulic / Mechanical Separation Actuator Specification	Specification
120	Mechanical Lock & Release Mechanism Design	Design
121	Manual Separation Lever Design & Human Factors	Design

No.	Document Title	Document Type
122	Separation Sequence Timing & Interlocks	Design
123	Separation Sensing & Confirmation Logic	Design
124	Clean-Break Electrical / Data / Fluid Disconnect Design	Design
125	Umbilical Design & Auto-Disconnect Mechanism	Design
126	Separation Dynamics Analysis	Analysis
127	Re-contact & Entanglement Risk Assessment	Safety
128	Separation System FMECA	Safety
129	Separation System Reliability Analysis	Analysis
130	Layer 1 – Primary Emergency Detection / Initiation Design	Design
131	Layer 2 – Independent Backup Initiation Path Design	Design
132	Layer 3 – Independent Manual / Crew Initiation Design	Design
133	Layer 4 – Independent Final / Last-Resort Separation Design	Design
134	Layer 5 – Direct-Impact Survival Design	Design
135	Layer 6 – Post-Event Survival and Recovery Design	Design
136	Safety-Layer Independence Analysis (Common-Cause Controlled)	Safety
137	Six-Layer Survival Architecture Detailed Specification	Architecture
138	Safety Layer Timing & Sequence Diagrams	Design
139	Crew-Commanded vs Automatic Activation Criteria	Requirements
140	Functional Hazard Assessment – Safety Layers	Safety
141	Preliminary System Safety Assessment – Safety Layers	Safety
142	Safety Requirements Allocation Matrix	Safety
143	Emergency Procedures and Crew Actions	Operations
144	Safety System Interface Control Document	ICD
145	Safety System Test and Verification Plan Outline	Test
146	Safety Case Structure and Argument Outline	Safety
147	Safety Family Risk Register	Safety
148	Independence Demonstration Plan (Evidence-Gated)	Safety
149	Safety Critical Design Review Checklist	Review
150	Safety System Certification Considerations	Certification
151	Multi-Mechanism Release Hardware Specification	Specification
152	Pyrotechnic Device Qualification Requirements	Requirements
153	Thermal Battery Specification & Life Assessment	Specification
154	Cold-Gas Bottle and Valve Specification	Specification
155	Safety Family Document Index and Traceability	Management
	GROUP F – Parachute, Landing Legs & Impact Energy Management (156 – 185)	
156	Parachute System Requirements (8 Primary + 8 Auxiliary)	Requirements
157	Parachute System Architecture and Deployment Sequence	Architecture
158	Primary Canopy Specification (Sizing Verification-Gated)	Specification
159	Auxiliary / Stabiliser Canopy Specification	Specification
160	Pilot / Extractor Chute Specification	Specification
161	Parachute Attachment and Load Path Design	Design
162	Parachute Inflation Confirmation Sensor Specification	Specification

No.	Document Title	Document Type
163	Parachute Deployment Logic and Failure Injection Cases	Design
164	Parachute System Mass and CG Contribution	Analysis
165	Parachute System FMECA	Safety
166	Landing System Requirements Document	Requirements
167	Landing Gear Architecture (8 Legs)	Architecture
168	Energy-Absorbing Landing Leg Specification	Specification
169	Progressive Hydraulic Damper Specification	Specification
170	Multi-Rate Elastomer Pad Specification	Specification
171	Landing Loads and Energy Absorption Analysis Plan	Analysis
172	Terrain Adaptation and Ground Pressure Requirements	Requirements
173	Crashworthiness Design Requirements	Requirements
174	Crash Load Cases and Occupant Protection Criteria	Safety
175	Energy-Absorbing Seat Specification	Specification
176	Restraint System Requirements	Requirements
177	Occupant Protection under Multi-Axis Loads	Analysis
178	Landing System Interface Control Document	ICD
179	Landing and Crashworthiness Test Plan Outline	Test
180	Drop Test Matrix and Instrumentation Requirements	Test
181	Landing System Mass and CG Contribution	Analysis
182	Landing Family Risk Assessment	Safety
183	Crashworthiness Certification Considerations	Certification
184	Landing Family Critical Design Review Checklist	Review
185	Parachute & Landing Family Document Index	Management
	GROUP G – Life Support, Fire, Flotation & Post-Event Survival (186 – 210)	
186	Life Support System Requirements (72-hour target)	Requirements
187	Life Support System Architecture (O■, CO■, Thermal)	Architecture
188	Oxygen System Design (Stored + Chemical)	Design
189	CO■ Scrubbing System Design	Design
190	Water and Rations Stowage Design	Design
191	Thermal Control and Insulation Design	Design
192	Life Support Endurance Analysis (Balance-Controlled)	Analysis
193	Life Support System FMECA	Safety
194	Flotation and Self-Righting System Concept	Design
195	Flotation Volume Sizing and Deployment Design	Design
196	Self-Righting Geometry and Stability Analysis	Analysis
197	Location Beacons and Recovery Aids Specification	Specification
198	406 MHz COSPAS-SARSAT Beacon Integration	Design
199	GNSS and Two-Way Satellite Messaging Design	Design
200	Visual Strobes and Acoustic Beacon Specification	Specification
201	Fire Protection System Specification (Hazard-Based)	Specification
202	Fire Detection and Suppression Logic	Design
203	Environmental Sealing and Contamination Control	Design

No.	Document Title	Document Type
204	Life Support / Environmental Interface Control Document	ICD
205	Life Support Test and Verification Plan Outline	Test
206	Flotation and Self-Righting Test Plan Outline	Test
207	Fire Protection Test Requirements	Test
208	Life Support Family Risk Assessment	Safety
209	Life Support Critical Design Review Checklist	Review
210	Life Support Family Document Index	Management
	GROUP H – Avionics, Electrical, Software & Common-Cause Analysis (211 – 235)	
211	Electrical Power System Architecture	Architecture
212	Emergency Battery Bus Design and Health Monitoring	Design
213	Backup Power Sources and Automatic Switch-over	Design
214	Electrical Power Distribution and Protection	Design
215	Avionics Architecture Description	Architecture
216	Sensor Suite Requirements and Redundancy	Requirements
217	Flight / Capsule Control Computer Specification	Specification
218	Human-Machine Interface Requirements	Requirements
219	Display and Alerting System Requirements	Requirements
220	Data Buses and Communication Architecture	Architecture
221	Navigation and Positioning System Requirements	Requirements
222	Communication System Requirements (Internal + External)	Requirements
223	Health Monitoring and Prognostics Concept	Design
224	Software Architecture and Partitioning	Software
225	Software Development Assurance Levels	Software
226	Software Requirements Specification – Safety Critical	Software
227	Electromagnetic Compatibility Requirements	Requirements
228	Cybersecurity for Flight-Critical and Safety Systems	Security
229	Avionics Environmental Qualification Requirements	Requirements
230	Electrical / Avionics Interface Control Document	ICD
231	Electrical / Avionics Mass and Power Budget	Analysis
232	Common-Cause Analysis – Electrical & Software	Safety
233	Avionics Family Risk Assessment	Safety
234	Avionics Critical Design Review Checklist	Review
235	Avionics Family Document Index	Management
	GROUP I – Human Factors, Egress, Verification Gates & Traceability (236 – 250)	
236	Human Factors Engineering Requirements	Requirements
237	Crew Station Layout and Ergonomics	Design
238	Seat Design and Energy-Absorbing Suspension	Design
239	Restraint System Design and Human Factors	Design
240	Egress Path Design (Side Doors + Roof Hatch)	Design
241	Egress Time and Unobstructed Path Analysis	Analysis
242	Internal Volume and Clearance Requirements	Requirements
243	Control Reach and Emergency Control Accessibility	Design

No.	Document Title	Document Type
244	Lighting, Marking and Emergency Wayfinding	Design
245	Noise, Vibration and Comfort Requirements	Requirements
246	Human Factors Test and Evaluation Plan Outline	Test
247	Occupant Injury Criteria and Assessment Methods	Safety
248	Crew Training and Emergency Procedure Outline	Operations
249	Master Verification Gate & Evidence Status Register	Management
250	Human Factors & Traceability Family Document Index	Management

This Master Document Tree (Rev 1.1) is controlled against Document 00 Version 3.0 — Comprehensive Engineering Baseline. All numerical values, architectural decisions and safety principles in the 250 documents shall remain consistent with the locked baseline. Performance claims remain verification-gated. Generation of individual documents shall prioritise critical-path families: Safety Layers & Common-Cause (Group E), Structures & Roll-Cage (Group D), and Requirements / Safety Case (Group B). No silent baseline changes are permitted.

— End of Master Document Tree – 250 Engineering Documents (Rev 1.1 – Aligned to Doc 00 V3.0) —
Ultra-Safe Central Capsule | Document 00 Version 3.0 | 5.80 m × 3.80 m | 22 t Hard Ceiling | 8 Legs